FitzCLIM

A Common Lisp Implementation of the CLIM II Protocols and the CLIM 2.2 Revisions

https://github.com/cgore/FitzCLIM/

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Introduction

FitzCLIM is a Common Lisp implementation of the CLIM II protocols [4] and of the CLIM 2.2 revisions. The 2.2 revisions are recorded in Franz's CLIM 2.2 User Guide [1], not in a second formal specification. That guide is copyrighted. This manual does not copy it. FitzCLIM is aimed at SBCL.

The frame manager realizes the standard gadgets as native controls. On Windows those are Windows controls. On macOS those are Cocoa controls. Layout positions them through the sheet protocol. A native control does not hold output records. Its items may still be Lisp objects, and a selection may run a command.

An application pane is drawn by FitzCLIM. It records output and holds presentations. A command loop reads commands, from typing or from a gesture on a presentation, and runs them.

A 3D pane is a native OpenGL child window, a sibling of the native controls. Pointer events at its edge become a ray. A hit returns a Lisp object and enters the same command and translator machinery as a 2D presentation. The OpenGL pane owns its redraw. FitzCLIM may ask that pane to repaint. The pane does not record its scene as output records. A label in the scene is drawn and picked there.

The 2D graphics protocol stays two-dimensional. Regions, transforms, clipping, formatted output, and output recording are a plane. They stay out of the scene graph.

FitzCLIM was begun so that EDEFB [2], the Elite Dangerous Electronic Flight Bag, can have a CLIM. The toolkit is general. Other programs will use it too. Elite Dangerous stays in the EDEFB repository.

This is a clean-room implementation. It is not derived from McCLIM, from Franz CLIM, or from LispWorks CLIM. The specification may be consulted. Its text stays out of this tree.

Tests use Σ [3], in particular the `behavior` and `should` forms in `sigma/behave`.

The frame managers target Windows and macOS. Geometry, ink, sheets, commands, frames, events, presentations, output recording, gadgets, pane layout, and the frame managers are under `source/`. An application pane records what it draws and finds the presentation under a pointer. A frame can stack its panes. A gadget holds a value or

runs a callback. On Windows the frame manager realizes that frame as a window and those gadgets as native controls. Pointer and keyboard input on an application pane become CLIM events, and a paint replays the recorded output. On macOS the frame manager realizes the frame as a Cocoa window and the gadgets as Cocoa controls. Pointer and keyboard input become CLIM events, and a menu of the frame's commands is installed in the menu bar.

You can download the library from GitHub at:
`https://github.com/cgore/FitzCLIM`

Getting Lisp

Before using this library you need a working Lisp. I use and recommend SBCL, Steel Bank Common Lisp, which is available at:
`http://www.sbcl.org`

Getting EMACS and SLIME

The best way to interact with any Common Lisp is via SLIME, the Superior Lisp Interaction Mode for EMACS. It is available at:
`https://common-lisp.net/project/slime/`

Bibliography

[1] FRANZ INC. *CLIM 2.2 User Guide*, 2.2.2 ed., 2005. September 2005. Copyright Franz Inc. Consult; do not copy.

[2] GORE, C. M. *EDEFB: the Elite Dangerous Electronic Flight Bag*, 2026. `https://github.com/cgore/edefb`.

[3] GORE, C. M. *Σ: A Library for ANSI Common Lisp*, 2026. `https://github.com/cgore/sigma`.

[4] MCKAY, S., AND YORK, W. *Common Lisp Interface Manager: CLIM II Specification*, 1994.